

Read Me First

Before opening any other file, read this document completely. It will save you hours of confusion and ensure you get maximum value from your purchase immediately.



Important: Read This Before Installation

OQUL EdTech OS is a full-stack production application, not a simple template. It requires a proper development environment. Please complete this document before attempting installation.

What Is OQUL?

OQUL (عقل — Arabic for *mind/intellect*) is a complete AI-powered educational platform originally built for Moroccan K–12 students. It features an intelligent AI tutor named **Leila**, adaptive learning paths, a full curriculum registry aligned to official educational standards, and a parent monitoring portal.

As a commercial product, OQUL gives you a **fully-built, production-tested SaaS foundation** that you can deploy, customize, and operate as your own educational business.

System Requirements

Requirement	Minimum	Recommended
Node.js	v20.0+	v20 LTS
PostgreSQL	v14+	v16
Redis	v7+	v7.2
RAM	4 GB	8 GB+
Storage	10 GB	50 GB+
OS	Ubuntu 22.04+	Ubuntu 22.04 LTS
Docker	v24+	Docker Desktop latest

Required API Keys

Groq API Required

Powers the Leila AI tutor with ultra-fast LLM inference. Free tier available at console.groq.com

SMTP Service Optional

For email verification and password resets. Works with Resend, SendGrid, or any SMTP provider.

Folder	Contents
00_START_HERE	Cover, Welcome, this document, and Folder Guide
01_INSTALLATION	Step-by-step setup guide and Quick Start
02_DOCUMENTATION	Business Overview, Architecture, SOP Library
03_TEMPLATES	Notion, Google Sheets, Canva, Word templates
04_LEGAL	License and Terms of Use
05_SUPPORT	FAQ and Support Guide
06_META	Version History, Changelog, Asset Manifest, Roadmap
source/	Complete application source code

Technology Stack at a Glance

Frontend → Next.js 15.5 → React 19 → TypeScript → Tailwind CSS → Radix UI	Backend → Node.js 20+ → PostgreSQL → Redis + BullMQ → Drizzle ORM → Groq AI API	DevOps → Docker Compose → GitHub Actions → Prometheus → Grafana → K6 Load Tests
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Key Concepts to Understand

| Leila — The AI Tutor

Leila is the core AI personality of the platform. She communicates in Arabic, French, and Moroccan Darija dialect. She adapts her teaching style based on the student's confidence level, pace, and learning history. Leila's persona is defined in `src/server/ai/personas/`.

| Adaptive Learning Engine

The platform continuously adjusts difficulty, pacing, and content based on student performance. This is powered by a knowledge tracing algorithm in `src/server/adaptive/` and a confidence engine in `src/server/confidence-engine/`.

| Curriculum Registry

All curriculum data lives in `curriculum-registry/` as structured JSON files. These map to official Moroccan educational standards and are seeded into the database on first run via scripts in `scripts/`.



Ready to Install?

Once you have Node.js 20+, PostgreSQL, Redis, and your Groq API key ready, open **01_INSTALLATION/Installation_Guide.pdf** for the complete step-by-step walkthrough.